

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Examination, 2016-17

Paper No. MTB-302: Differential Equations (New & Old Course)

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) If y_1, y_2, y_3 are solutions of $y'' + ay' + by = 0$, show that the Wronskian $W(y_1, y_2, y_3) = 0$.
- (b) Show that the number of integrating factors of $M(x, y) dx + N(x, y) dy = 0$, which has a solution, is infinite.
- (c) Find the general solution of $p^2 + p - 6 = 0$, where $p = \frac{dy}{dx}$.
- (d) Define singular solution of a differential equation.
- (e) Obtain the linear homogeneous 2nd order ODE whose linearly independent solutions are $y = x$ and $y = x^2$.
- (f) Transform $x^2 \frac{d^2 y}{dx^2} + ax \frac{dy}{dx} + by = 0$ by substituting $x = e^t$.
- (g) Obtain the condition for $y = e^{mx}$ to be a solution of $\frac{d^2 y}{dx^2} + c \frac{dy}{dx} + Py = 0$.
- (h) Examine whether $\sin x \frac{dy}{dx} + (\cos x + \sin x) \frac{dy}{dx} + y \cos x = 0$ is exact.
- (i) Define regular singular points for a 2nd-order linear ODE and give an example.
- (j) Write the value of $P_n(0)$ for $n = 0, 1, 2, \dots$.
- (k) Show that $J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x$.

Question 2.

- (a) Show that the ODE $M dx + N dy = 0$ (where M, N have continuous partial derivatives) is exact if and only if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$. [6]
- (b) Solve: $(x + y + 1) dx = (2x + 2y + 1) dy$. [6]

Question 3.

(a) Solve $\frac{d^2y}{dx^2} + y = \cos x$ by the method of variation of parameters. [6]

(b) Solve $x^2 \frac{d^2y}{dx^2} - 2x(3x-2) \frac{dy}{dx} + 3x(3x-4)y = e^{3x}$ by reducing to normal form. [6]

Question 4.

(a) Find the general solution of n th-order linear homogeneous ODE with constant coefficients when the auxiliary equation has real and distinct roots. [6]

(b) Find the solution of $\frac{d^3y}{dx^3} - 5\frac{d^2y}{dx^2} + \frac{dy}{dx} + y = xe^x + e^x$ satisfying $y = 0$, $\frac{dy}{dx} = 0$, $\frac{d^2y}{dx^2} = 0$ at $x = 0$. [6]

Question 5.

(a) Reduce $(px^2 + y^2)(px + y) = (p+1)^2$ (where $p = dy/dx$) to Clairaut's form using $u = xy$, $v = x + y$ and find the complete primitive. [6]

(b) Find the singular solution of $\sin\left(x \frac{dy}{dx}\right) \cos y - \cos\left(x \frac{dy}{dx}\right) \sin y = 0$. [6]

Question 6.

(a) Solve $\frac{d^2y}{dx^2} + (x^2 - 1) \frac{dy}{dx} + y = (e^x - 1)y$ in series about the ordinary point $x = 0$. [6]

(b) Express $f(x) = x^4 + 3x^3 - x^2 + 5x - 2$ in terms of Legendre polynomials. [6]

Question 7.

(a) Show that $\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$. [6]

(b) Prove that $P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$, where $P_n(x)$ is the Legendre polynomial of degree n . [6]

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